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**PimpriChinchwad College of
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Sector No. 26, Pradhikaran,
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Internal Evaluation-II

Subject: “Digital Signal Processing Lab”

Project Title:

Designing a filter circuit to separate the voice signal and instrument signal from an audio song.

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Abstract

This report explores the application of Fast Fourier Transform (FFT) and digital filter design techniques for separating vocal and instrument signals in an audio song using MATLAB. The project aims to develop a digital filter circuit that effectively isolates these components based on their frequency characteristics. The report details the objectives, methodology, results obtained, and potential applications of this approach.

Introduction

Music recordings often contain a combination of vocals and instrumental elements. Separating these signals can be beneficial for various applications, including music remixing, audio restoration, and speech recognition. This project investigates the use of FFT and digital filter design in MATLAB to achieve audio source separation.

Objectives

1. To develop a high-quality FFT based audio separation algorithm in MATLAB for analyzing the frequency spectrum of an audio song.
2. To identify the typical frequency ranges occupied by vocals and instruments in the audio spectrum.
3. To design digital filters that effectively separate the vocal and instrument signals based on their frequency ranges.
4. To reconstruct the separated vocal and instrument signals using inverse FFT.

Methodology

The methodology for achieving the specified objectives involves the following steps

1. Data Acquisition: An audio song containing both vocals and instruments is selected as the input for the project.
2. FFT Implementation: An FFT algorithm is implemented in MATLAB to compute the frequency spectrum of the audio signal.
3. Frequency Analysis: The frequency spectrum is analyzed to identify the specific ranges typically occupied by vocals and instruments.
4. Filter Design: Digital filters are designed in MATLAB to pass the desired frequency range (vocals or instruments) while attenuating the undesired range.
5. Signal Separation: The audio signal is filtered using the designed filters to isolate the vocal and instrument components.
6. Signal Reconstruction: Inverse FFT is applied to the filtered signals to reconstruct the separated vocal and instrument waveforms.

Algorithm:

1. Load Audio: Load AudioSong.wav and normalize the audio signal.
2. Plot Original Signal: Plot the original audio signal.
3. High-Pass Filter: Design and apply a high-pass filter (1000 Hz) to separate instrumental from voice.
4. Plot Filtered Signal: Plot the filtered instrumental signal.
5. Calculate Voice Signal: Obtain the voice signal by subtracting the filtered from the original.
6. Plot Voice Signal: Plot the separated voice signal.
7. Plot FFTs:
 - FFT of original audio signal.
 - FFT of filtered instrumental signal.
 - FFT of voice signal.

MATLAB Code:

% Load the audio file

```
[y, Fs] = audioread('AudioSong.wav');
```

% Normalize the audio

```
y = y / max(abs(y));
```

% Define the length of the audio

```
N = length(y);
```

% Create time vector

```
t = (0:N-1) / Fs;
```

% Plot the original audio signal

```
figure;
```

```
subplot(4,1,1);
```

```
plot(t, y);
```

```
title('Original Audio Signal');
```

```
xlabel('Time (s)');
```

```
ylabel('Amplitude');
```

% Design a High-Pass Filter

```
fc = 1000; % Cutoff frequency in Hz
```

```
[b, a] = butter(6, fc/(Fs/2), 'high'); % 6th order Butterworth high-pass filter
```

% Apply the filter

```
filtered_signal = filter(b, a, y);
```

```

% Plot the filtered signal (instrumental)
subplot(4,1,2);
plot(t, filtered_signal);
title('Filtered Signal (Instrumental)');
xlabel('Time (s)');
ylabel('Amplitude');

% Calculate the voice (difference between original and filtered)
voice_signal = y - filtered_signal;

% Plot the voice signal
subplot(4,1,3);
plot(t, voice_signal);
title('Voice Signal');
xlabel('Time (s)');
ylabel('Amplitude');

% Compute FFT for original signal
Y = fft(y);
f = (0:N-1)*(Fs/N);
Y_mag = abs(Y);

% Plot FFT of original signal
subplot(4,1,4);
plot(f, Y_mag);
title('FFT of Original Audio Signal');
xlabel('Frequency (Hz)');
ylabel('Magnitude');

% Display the plots
sgtitle('Audio Signal Separation with FFT');

% Compute FFT for filtered instrumental signal
Filtered_signal_fft = fft(filtered_signal);
Filtered_signal_mag = abs(Filtered_signal_fft);

% Plot FFT of filtered instrumental signal
figure;
subplot(2,1,1);
plot(f, Filtered_signal_mag);
title('FFT of Filtered Signal (Instrumental)');

```

```

xlabel('Frequency (Hz)');
ylabel('Magnitude');

% Compute FFT for voice signal
Voice_signal_fft = fft(voice_signal);
Voice_signal_mag = abs(Voice_signal_fft);

```

```

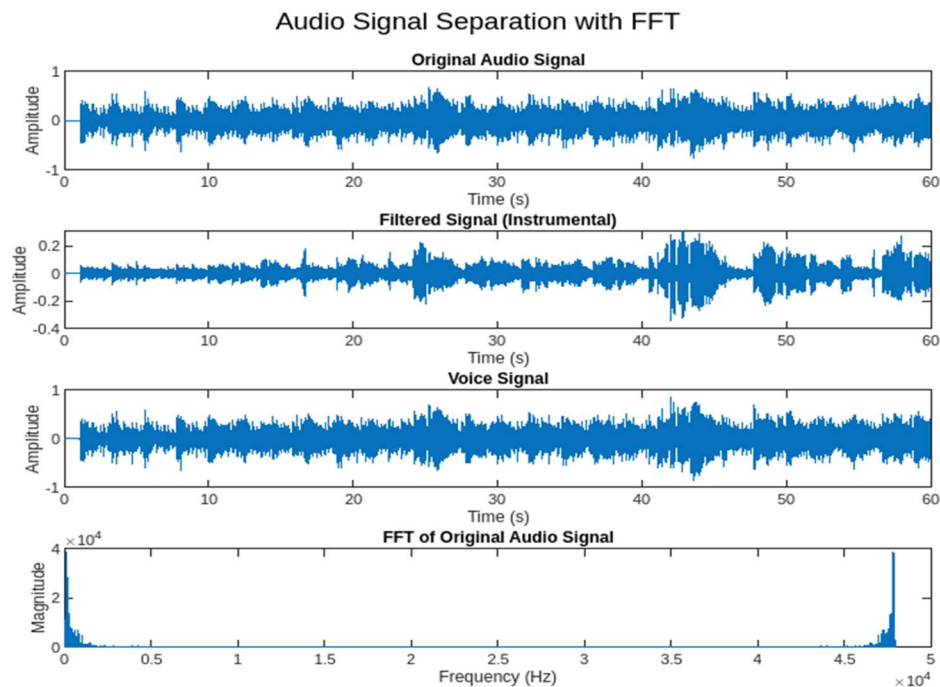
% Plot FFT of voice signal
subplot(2,1,2);
plot(f, Voice_signal_mag);
title('FFT of Voice Signal');
xlabel('Frequency (Hz)');
ylabel('Magnitude');

```

Results

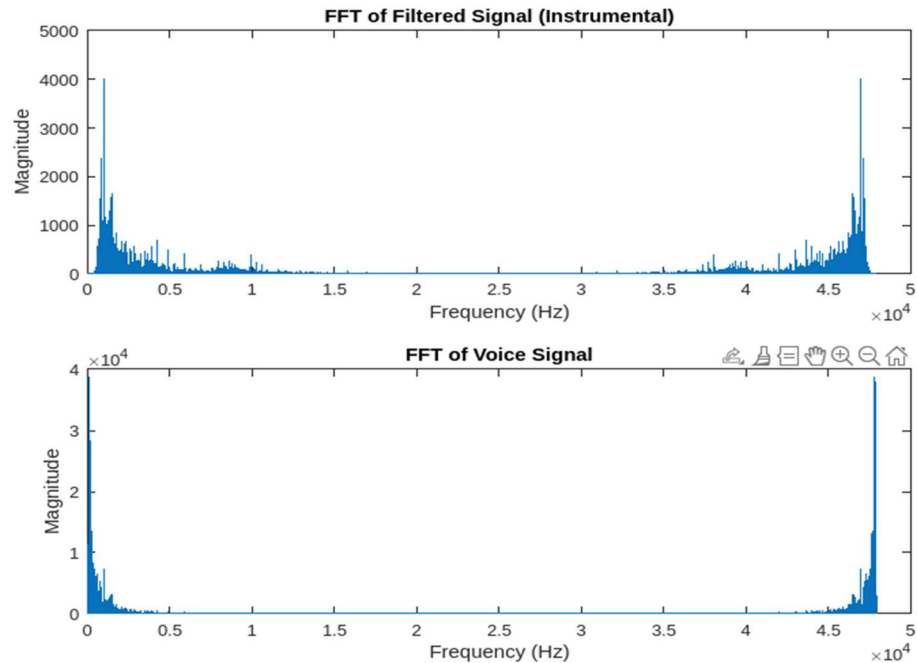
Time domain analysis

Time domain analysis is all about examining a signal's behavior over time. It focuses on the direct observation of how a signal's value changes with respect to time. Imagine a graph where the x-axis represents time and the y-axis represents the signal's strength or amplitude. Time domain analysis dissects the signal's characteristics directly from this graph.



Frequency Domain analysis

Frequency domain analysis is a technique used to understand how a signal is composed of different frequencies. It's a complementary approach to time domain analysis, which focuses on how a signal changes over time.



Conclusion

This project demonstrates the feasibility of using FFT and digital filter design techniques for audio source separation in MATLAB. By analyzing the frequency spectrum and applying appropriate filters, it is possible to effectively isolate vocal and instrument components from an audio signal. This technology holds promise for various applications in music production, audio restoration, and speech processing.